

Dataset	Method $\mathcal{M}$	$\text{Acc}(\mathcal{M}, \mathcal{Q}, \emptyset)$	$\text{Acc}(\mathcal{M}, \mathcal{Q}, \mathcal{P}_{\text{clean}})$	$\text{Acc}(\mathcal{M}, \mathcal{Q}, \mathcal{P}_{\text{Irrelevant}})$ (NDA)				$\text{Acc}(\mathcal{M}, \mathcal{Q}, \mathcal{P}_{\text{Inaccurate}})$ (NDA)			
				Easy	Medium	Hard	Avg.	Easy	Medium	Hard	Avg.
Math Base-9	Base	7.2	46.4	39.3 (81.9)	30.3 (58.9)	26.6 (49.5)	32.1 (63.5)	23.2 (40.8)	10.1 (7.4)	6.0 (-3.0)	13.1 (15.1)
	w/ ISC	8.7	24.3	17.7 (57.7)	14.7 (38.5)	12.7 (25.6)	15.0 (40.4)	18.4 (62.2)	13.7 (32.1)	12.3 (23.1)	14.8 (39.1)
	w/ SP	7.2	26.2	25.5 (96.3)	25.5 (96.3)	21.9 (77.4)	24.3 (90.0)	20.0 (67.4)	18.4 (58.9)	14.3 (37.4)	17.6 (54.7)
	w/ SM	12.3	37.4	30.0 (70.5)	22.7 (41.4)	16.5 (16.7)	23.1 (43.0)	24.7 (49.4)	19.2 (27.5)	12.4 (0.4)	18.8 (25.9)
	w/ SD	12.3	47.9	37.2 (69.9)	25.4 (36.8)	24.7 (34.8)	29.1 (47.1)	29.3 (47.8)	12.5 (0.6)	8.7 (-10.1)	16.8 (12.6)
	w/ SC	12.3	61.5	51.1 (78.9)	39.0 (48.6)	36.2 (54.3)	42.1 (60.6)	32.7 (41.5)	15.3 (6.1)	7.5 (-9.8)	18.5 (12.6)
	w/ SCO	12.5	53.6	46.3 (82.2)	39.6 (65.9)	36.4 (58.2)	40.8 (68.9)	34.7 (54.0)	22.0 (23.1)	17.7 (12.7)	24.8 (29.9)
	w/ BT	7.2	47.2	39.2 (80.0)	34.2 (67.5)	29.9 (56.8)	34.4 (68.0)	30.1 (57.2)	18.4 (28.0)	14.1 (17.2)	20.9 (34.2)
	w/ CC	7.2	44.9	43.3 (95.8)	44.6 (99.2)	45.5 (101.6)	44.5 (98.9)	37.2 (79.4)	31.7 (64.8)	30.7 (62.2)	33.2 (68.8)
	w/ CD-CoT	12.3	60.7	59.7 (97.9)	60.7 (100.0)	57.2 (92.8)	59.2 (96.9)	54.0 (86.2)	58.7 (95.9)	48.4 (74.6)	53.7 (85.5)
Math Base-11	Base	5.5	23.9	19.1 (73.9)	13.6 (44.0)	10.7 (28.3)	14.5 (48.9)	14.0 (46.2)	6.7 (6.5)	3.6 (-10.3)	8.1 (14.1)
	w/ ISC	7.4	11.2	8.3 (23.7)	7.8 (10.5)	6.0 (-36.8)	7.4 (0.0)	6.5 (-23.7)	5.2 (-57.9)	4.7 (-71.1)	5.5 (-50.0)
	w/ SP	5.5	20.7	17.5 (78.9)	16.7 (73.7)	14.0 (55.9)	16.0 (69.1)	14.1 (56.6)	10.7 (34.2)	10.8 (34.9)	11.9 (42.1)
	w/ SM	8.0	16.3	12.0 (48.2)	6.0 (-24.1)	5.7 (-27.7)	7.9 (-1.2)	12.0 (48.2)	9.3 (15.7)	7.7 (-3.6)	9.7 (20.5)
	w/ SD	8.0	17.9	12.3 (43.4)	12.0 (40.4)	13.3 (53.5)	12.5 (45.5)	17.0 (90.9)	8.7 (7.1)	5.3 (-27.3)	10.3 (23.2)
	w/ SC	8.0	33.7	25.3 (67.3)	16.3 (32.3)	15.0 (27.2)	18.9 (42.4)	19.7 (45.5)	9.3 (5.1)	3.3 (-18.3)	10.8 (10.9)
	w/ SCO	22.9	33.0	29.2 (62.4)	24.0 (10.9)	20.0 (-28.7)	24.4 (14.9)	29.2 (62.4)	20.0 (-28.7)	17.2 (-56.4)	22.1 (-7.9)
	w/ BT	5.5	24.3	17.9 (66.0)	17.2 (62.2)	13.7 (43.6)	16.3 (57.4)	12.8 (38.8)	9.2 (19.7)	6.8 (6.9)	9.6 (21.8)
	w/ CC	5.5	22.3	19.1 (81.0)	18.4 (76.8)	18.2 (75.6)	18.6 (78.0)	19.0 (80.4)	15.3 (58.3)	14.6 (54.2)	16.3 (64.3)
	w/ CD-CoT	8.0	31.0	33.7 (111.7)	32.7 (107.4)	34.7 (116.1)	33.7 (111.7)	29.0 (91.3)	30.7 (98.7)	25.3 (75.2)	28.3 (88.3)
Symbolic Equal	Base	8.8	32.7	28.1 (80.8)	25.1 (68.2)	23.0 (59.4)	25.4 (69.5)	29.1 (84.9)	26.1 (72.4)	22.7 (58.2)	26.0 (72.0)
	w/ ISC	5.7	23.9	20.0 (78.6)	16.3 (58.2)	15.5 (53.8)	17.3 (63.7)	19.2 (74.2)	18.3 (69.2)	18.1 (68.1)	18.5 (70.3)
	w/ SP	8.8	23.2	23.0 (98.6)	22.6 (95.8)	22.7 (96.5)	22.8 (97.2)	23.7 (103.5)	22.5 (95.1)	23.5 (102.1)	23.2 (100.0)
	w/ SM	9.7	25.0	20.7 (71.9)	19.7 (65.4)	16.7 (45.8)	19.0 (60.8)	21.0 (73.9)	20.3 (69.3)	20.0 (67.3)	20.4 (69.9)
	w/ SD	9.7	9.9	10.1 (—)	10.9 (—)	10.3 (—)	10.4 (—)	10.1 (—)	10.9 (—)	10.4 (—)	10.5 (—)
	w/ SC	9.7	35.3	31.0 (83.2)	28.3 (72.7)	27.0 (67.6)	28.8 (74.6)	33.3 (92.2)	30.7 (82.0)	26.0 (63.7)	30.0 (79.3)
	w/ SCO	12.5	38.5	34.9 (86.2)	33.4 (80.4)	32.7 (77.7)	33.7 (81.5)	34.0 (82.7)	34.1 (83.1)	34.5 (84.6)	34.2 (83.5)
	w/ BT	8.8	31.8	26.0 (74.8)	22.7 (60.4)	22.6 (60.0)	23.8 (65.2)	26.3 (76.1)	22.7 (60.4)	22.9 (61.3)	24.0 (66.1)
	w/ CC	8.8	37.8	33.8 (86.2)	32.7 (82.4)	32.0 (80.0)	32.8 (82.8)	31.3 (77.6)	33.0 (83.4)	29.9 (72.8)	31.4 (77.9)
	w/ CD-CoT	9.7	42.7	44.7 (106.1)	42.7 (100.0)	44.0 (103.9)	43.8 (103.3)	42.6 (99.7)	41.3 (95.8)	42.7 (100.0)	42.2 (98.5)
Symbolic Longer	Base	0.0	9.2	6.3 (68.5)	7.2 (78.3)	6.0 (65.2)	6.5 (70.7)	7.0 (76.1)	6.8 (73.9)	6.0 (65.2)	6.6 (71.7)
	w/ ISC	0.1	4.9	4.6 (93.7)	2.7 (54.2)	3.7 (75.0)	3.7 (75.0)	3.4 (68.7)	4.3 (87.5)	3.3 (66.7)	3.7 (75.0)
	w/ SP	0.0	5.1	4.3 (84.3)	4.1 (80.4)	3.9 (76.5)	4.1 (80.4)	4.9 (96.1)	4.0 (78.4)	4.5 (88.2)	4.5 (88.2)
	w/ SM	0.0	1.7	0.7 (—)	0.7 (—)	1.3 (—)	1.0 (—)	1.3 (—)	0.7 (—)	0.3 (—)	0.8 (—)
	w/ SD	0.0	0.1	0.1 (—)	0.1 (—)	0.2 (—)	0.1 (—)	0.1 (—)	0.3 (—)	0.0 (—)	0.1 (—)
	w/ SC	0.0	13.0	7.7 (59.2)	9.0 (69.2)	6.3 (48.5)	7.7 (59.2)	8.0 (61.5)	8.0 (61.5)	8.7 (66.9)	8.2 (63.1)
	w/ SCO	0.0	18.7	12.1 (64.7)	10.5 (56.1)	11.3 (60.4)	11.3 (60.4)	15.2 (81.3)	15.9 (85.0)	9.8 (52.4)	13.6 (72.7)
	w/ BT	0.0	7.2	3.4 (47.2)	3.5 (48.6)	2.5 (34.7)	3.1 (43.1)	3.8 (52.8)	3.6 (50.0)	3.6 (50.0)	3.7 (51.4)
	w/ CC	0.0	9.4	9.8 (104.3)	7.9 (84.0)	7.9 (84.0)	8.5 (90.4)	8.5 (90.4)	7.4 (78.7)	6.5 (69.1)	7.5 (79.8)
	w/ CD-CoT	0.0	12.3	<u>12.0 (97.6)</u>	12.0 (97.6)	13.0 (105.7)	12.3 (100.0)	<u>12.3 (100.0)</u>	<u>10.0 (81.3)</u>	11.0 (89.4)	<u>11.1 (90.2)</u>

Table 32: Comparing accuracy and NDA metrics across all methods. The **boldface** numbers mean the best results; underlines indicate the second-best. "—" denotes methods with poor results, where zero-shot performance approximates clean 3-shot outcomes, rendering NDA calculation meaningless.

Method $\mathcal{M}$	Clean ($\text{Acc } \mu \pm \sigma$)	Irrelevant ($\text{Acc } \mu \pm \sigma$)			Inaccurate ($\text{Acc } \mu \pm \sigma$)		
		Easy	Medium	Hard	Easy	Medium	Hard
Base	46.4 ± 2.54	39.3 ± 2.22	30.3 ± 2.62	26.6 ± 2.17	23.2 ± 2.92	10.1 ± 1.34	6.0 ± 1.42
w/ ISC	24.3 ± 2.00	17.7 ± 1.72	14.7 ± 1.04	12.7 ± 1.19	18.4 ± 1.75	13.7 ± 1.22	12.3 ± 0.82
w/ SP	26.2 ± 2.06	25.5 ± 0.87	25.5 ± 1.48	21.9 ± 0.69	20.0 ± 1.65	18.4 ± 1.32	14.3 ± 2.17
w/ SM	37.4 ± 6.58	30.0 ± 3.82	22.7 ± 4.24	16.5 ± 7.77	24.7 ± 3.32	19.2 ± 2.62	12.4 ± 1.91
w/ SD	47.9 ± 1.63	37.2 ± 0.71	25.4 ± 3.68	24.7 ± 1.41	29.3 ± 0.58	12.5 ± 0.28	8.7 ± 2.33
w/ SC	61.5 ± 0.68	51.1 ± 1.25	39.0 ± 2.14	36.2 ± 2.31	32.7 ± 1.19	15.3 ± 0.91	7.5 ± 1.37
w/ SCO	53.6 ± 0.72	46.3 ± 1.60	39.6 ± 1.61	36.4 ± 0.15	34.7 ± 0.88	22.0 ± 1.18	17.7 ± 1.99
w/ BT	47.2 ± 1.35	39.2 ± 1.35	34.2 ± 2.74	29.9 ± 2.10	30.1 ± 1.59	18.4 ± 2.03	14.1 ± 2.90
w/ CC	44.9 ± 3.80	43.3 ± 1.36	44.6 ± 3.23	45.5 ± 2.88	37.2 ± 1.85	31.7 ± 2.15	30.7 ± 2.94
w/ CD-CoT	60.7 ± 0.49	59.7 ± 0.49	60.7 ± 3.18	57.2 ± 1.63	54.0 ± 1.41	58.7 ± 4.53	48.4 ± 1.91

Table 33: Comparing the mean values and standard deviation values of accuracy across all methods.

The two important aspects of the proposed method are (1) contrastive denoising with rationale rephrasing (+ CD Rephrase) and (2) repeated reasoning with voting (+ Vote). Here, we conduct an extra ablation study on the NoRa-Math Base-9 dataset to justify the denoising power of CD-CoT.

In this ablation study, we compare the performance of CD-CoT with different combinations of clean and noisy rationales on the NoRa-Math Base-9 dataset. The "Base model" column represents the accuracy of the base model without any additional components. The "+ CD Rephrase" column shows the accuracy when the contrastive denoising with rationale rephrasing (CD Rephrase) is applied. The "+ Vote" column represents the accuracy when the final answer is determined by a majority vote of five runs. Finally, the "CD-CoT" column shows the accuracy of the complete CD-CoT method, which includes both CD rephrasing and voting.

Since CD-CoT requires 3-shot noisy rationales and 1-shot clean rationale for contrastive rephrasing, we compare its performance with the following setups. The results are summarized in the Tab. 34.

- 3-shot clean rationales (without CD-CoT)
- 3-shot noisy rationales (irrelevant or inaccurate) of varying difficulty levels + 1-shot clean rationale (with CD-CoT)
- 4-shot clean rationales (without CD-CoT)

CoT shots	Base model	+ CD Rephrase	+ Vote	CD-CoT
3-shot clean shots	46.4	—	62.3	—
3-shot irrelevant-easy + 1-shot clean	46.0	47.2	57.3	60.0
3-shot irrelevant-medium + 1-shot clean	41.3	47.2	54.7	60.7
3-shot irrelevant-hard + 1-shot clean	36.2	46.5	47.3	58.3
3-shot inaccurate-easy + 1-shot clean	36.5	42.6	44.7	53.0
3-shot inaccurate-medium + 1-shot clean	23.4	43.0	32.3	58.7
3-shot inaccurate-hard + 1-shot clean	16.1	37.0	24.3	47.0
4-shot clean rationale	50.8	—	67.3	—

Table 34: Ablation study of CD rephrase and vote components on the NoRa-Math Base-9 dataset.

From the experimental results, we can observe that:

Observation F.1 (CD-CoT significantly improves reasoning performance across various noisy scenarios). CD-CoT brings a significant improvement compared to the base model when prompted with 3 noisy rationales + 1 clean rationale across various noise ratios of both irrelevant and inaccurate noise. The base accuracy of CD-CoT can be attributed to the improvement brought by the contrastive denoising with rationale rephrasing method. This accuracy is further enhanced in the voting method, which demonstrates the effectiveness of the self-consistency approach in improving reasoning performance.

Observation F.2 (Noisy rationales with CD-CoT does not outperform full clean rationales). The accuracy before voting and the accuracy after voting in CD-CoT do not exceed the accuracy of the base model prompted with 4 clean rationales. This indicates that while CD-CoT is effective at denoising rationales and enhancing reasoning performance in noisy situations, it does not inherently improve reasoning beyond the capabilities of the model when clean rationales are provided.

Observation F.3 (Contrastive denoising and voting in CD-CoT are important). The results highlight the importance and effectiveness of the two key aspects of CD-CoT: (1) contrastive denoising with rationale rephrasing, which helps to filter out the noise and align the format of the rationales, and (2) repeated reasoning with voting, which enhances the robustness and accuracy of the final predictions. Notably, CD-CoT proves to be a valuable approach for improving the reasoning performance in the presence of noisy rationales, though it does not surpass the performance with full clean rationales.

Observation F.4 (CD-CoT is more effective at managing irrelevant noise compared to inaccurate noise). As shown in Tab. 33, CD-CoT is more effective at handling irrelevant noise than it is at managing inaccurate noise at the same level. This is because irrelevant noise is easier to differentiate from the target information, making it simpler to remove during the rephrasing step of the contrasting denoising process. Additionally, Appendix. F.9 presents the denoised results for high-noise scenarios.

Variants of prompt for CD-CoT. We also conduct additional experiments to test the sensitivity of CD-CoT to prompt designs. Specifically, we generate several variants of prompts for the proposed CD-CoT method, including a shorter prompt, the default prompt, and a longer one:

The Shorter Prompt

Here are two examples: the first one has proper explanation and answer, while the second one has distracted explanation and correct answer. Please follow the first example’s explanation and provide the correct explanation and answer for the second one.

The Default Prompt

Here are two examples for the same type of task: the first example has correct explanation and correct answer, and the second example has distracted explanation and correct answer. Please follow the first example and give me the correct explanation and answer for the second example, which should be logically consistent with the first one.

The Longer Prompt

The following presents two examples of the same type of task. The first example contains both a correct explanation and a correct answer. The second example, however, includes a distracted explanation but still provides the correct answer. Your task is to analyze these examples and then provide a revised version explanation of the second example along with its answer. Ensure that your revised explanation is logically consistent with the first example.

We conduct experiments on NoRa-Math Base-9 to compare these three prompts. The results in Tab. 35 show that the performance of CD-CoT is only marginally influenced by these prompts. Note that the prompt of CD-CoT remains simple. Cooperating with advanced methods for iterating prompts and rationales, such as APE [110] and Star [97], can further improve this method.

method	Irrelevant-medium	Inaccurate-medium
Base Model	0.284	0.082
CD-CoT w/ original prompt	0.49	0.48
CD-CoT w/ short prompt	0.46	0.46
CD-CoT w/ long prompt	0.47	0.48

Table 35: Ablation study of different prompts in CD-CoT.

Rephrasing performance of SD. Besides, we observe that the SD method performs better on some tasks when using irrelevant and inaccurate rationales compared to clean ones. A thorough examination of the empirical results from the log files reveals some inconsistencies in the performance of the SD method, especially in the NoRa-Symbolic Equal and NoRa-Commonsense tasks.

Specifically, for NoRa-Symbolic Equal task:

- In this task, comparing SD and SC reveals that the domain knowledge of LLM is relatively weak. According to the experiment results, when SD rephrases symbolic tasks, it often directly returns the final answer of the original response in many cases.
- The quality of the rephrased results basically does not change with the increase in noise.
- As a result, the accuracies of SD under clean and noisy settings are very close, with a maximum difference of less than 1%, which we think is within the acceptable range of error.

For NoRa-Commonsense task:

- We observe different behaviors of SD in clean and noisy settings.
- In noisy settings, SD does not follow the prompt to make minor corrections. Instead, SD generates short rationales with only the answer, discarding the original rationale (as shown in Tables. 36-37). This simplification can benefit commonsense tasks where the model already has a strong inherent understanding.
- On the other hand, in the clean setting, SD unexpectedly generates entirely new rationales instead of making minor corrections. These new rationales may deviate from the original correct reasoning, potentially introducing errors or inconsistencies that weren't present in the initial, accurate solution.

In Tab. 36 and Tab. 37, we present two representative examples of SD in the commonsense task. Our findings show that: (i) when provided with a clear rationale, SD generates entirely new rationales rather than simply making minor corrections; (ii) when given a noisy rationale, SD produces brief rationales that include only the answer, completely discarding the original rationale.

Overall, SD's denoising effect relies on LLMs' intrinsic capability. SD performs five maskings and reconstructions per noisy example, concatenating results into five prompts for LLM inference. In contrast, SC performs five direct inferences on noisy prompts without explicit denoising. By comparing SD and SC's performance, we can gain insights into the model's denoising effects.

In Tab. 3, SC outperforms SD in NoRa-Math and NoRa-Symbolic tasks, while SD only marginally excels in the NoRa-Commonsense task. This pattern persists in the clean settings. This suggests that